

CBSE Class 12 Physics — Half-Yearly Predicted Paper (Set 4 of 10)

Section A: 1-Mark Questions (MCQ/Assertion-Reason)

Q1. A certain charge Q is divided into two parts, q and $Q - q$, which are then separated by a certain fixed distance. For the electrostatic force between them to be absolutely maximum, the ratio q/Q must be: (a) $1/2$ (b) $1/3$ (c) $1/4$ (d) $2/3$ [Practice] | **[Chapter 1: Electric Charges & Fields | Confidence: Very High]**

Q2. Two large parallel conducting plates are placed in a vacuum. A uniform electric field E exists between them. The work done in moving a point charge q parallel to the plates over a distance d is: (a) qEd (b) $qEd/2$ (c) Zero (d) $-qEd$ [PYQ 2020] | **[Chapter 2: Electrostatic Potential & Capacitance | Confidence: High]**

Q3. A uniform metallic wire of resistance R and resistivity ρ is stretched physically to double its original length. The new resistivity of the material of the wire will be: (a) 2ρ (b) 4ρ (c) $\rho/2$ (d) ρ [PYQ 2023] | **[Chapter 3: Current Electricity | Confidence: Very High]**

Q4. A circular coil of 1 turn and radius r carries a current I . The magnetic field at its centre is B . If the same wire is rewound to form a closely bound coil of n turns of a smaller radius and the same current I is passed through it, the new magnetic field at its centre will be: (a) nB (b) n^2B (c) B/n (d) B/n^2 [PYQ 2018] | **[Chapter 4: Moving Charges & Magnetism | Confidence: High]**

Q5. When a small piece of a diamagnetic substance is placed in a non-uniform magnetic field, it tends to move: (a) from the weaker part of the field to the stronger part. (b) from the stronger part of the field to the weaker part. (c) perpendicular to the magnetic field. (d) in a circular path. [PYQ 2016] | **[Chapter 5: Magnetism & Matter | Confidence: Very High]**

Q6. A metallic coin is dropped from rest such that it falls vertically through a strong, uniform horizontal magnetic field. The downward acceleration of the coin will be: (a) equal to g (b) strictly greater than g (c) strictly less than g (d) zero [PYQ 2020] | **[Chapter 6: Electromagnetic Induction | Confidence: High]**

Q7. What is the exact value of the power factor for a purely capacitive AC circuit? (a) 1 (b) 0.5 (c) 0 (d) Infinity [PYQ 2022] | **[Chapter 7: Alternating Current | Confidence: Very High]**

Q8. In an electromagnetic wave propagating in a vacuum, the ratio of the amplitude of the electric field (E_0) to the amplitude of the magnetic field (B_0) is exactly equal to: (a) the wavelength of the EM wave. (b) the frequency of the EM wave. (c) the speed of light in a vacuum (c). (d) the intrinsic impedance of free space. [PYQ 2015] | **[Chapter 8: Electromagnetic Waves | Confidence: Very High]**

Q9. A thin lens has an optical power of -4.0 D. The focal length and the inherent nature of the lens are respectively: (a) -25 cm, Convex (b) $+25$ cm, Concave (c) -25 cm, Concave (d) $+40$ cm, Convex [Practice] | **[Chapter 9: Ray Optics | Confidence: High]**

Q10. In a single slit diffraction experiment, the angular position of the very first minimum on either side of the central maximum for a slit of width a illuminated by light of wavelength λ is given by: (a) $\lambda/2a$ (b) λ/a (c) $2\lambda/a$ (d) $3\lambda/2a$ [PYQ 2021] | **[Chapter 10: Wave Optics | Confidence: Very High]**

Q11. The graph between the maximum kinetic energy (K_{max}) of emitted photoelectrons and the frequency (ν) of the incident radiation is a straight line. The slope of this straight line physically represents: (a) the Work function of the metal. (b) the Threshold frequency. (c) Planck's constant (h). (d) the Charge of an electron (e). [PYQ 2019] | **[Chapter 11: Dual Nature of Radiation & Matter | Confidence: Very High]**

Q12. If the kinetic energy of a free electron is suddenly increased by a factor of exactly 4, the de-Broglie wavelength associated with it will become: (a) 2 times its initial value. (b) $1/2$ of its initial value. (c) $1/4$ of its initial value. (d) 4 times its initial value. [PYQ 2020] | **[Chapter 11: Dual Nature of Radiation & Matter | Confidence: High]**

For Questions 13 to 16, two statements are given — one labelled Assertion (A) and the other labelled Reason (R). Select the correct answer from the codes (a), (b), (c), and (d) as given below. (a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is NOT the correct explanation of A. (c) A is true but R is false. (d) A is false and R is also false.

Q13. Assertion (A): The total electric flux passing through any closed Gaussian surface completely enclosing an electric dipole is always zero. **Reason (R):** The net electric charge of an electric dipole is identically zero. [PYQ 2023] | **[Chapter 1: Electric Charges & Fields | Confidence: Very High]**

Q14. Assertion (A): The terminal voltage of a cell delivering current to an external circuit is strictly less than its electromotive force (emf). **Reason (R):** A finite potential drop occurs across the internal resistance of the cell when a current flows through it. [PYQ 2022] | **[Chapter 3: Current Electricity | Confidence: Very High]**

Q15. Assertion (A): The magnitude of the induced emf in a coil depends explicitly on the rate of change of magnetic flux linked with it. **Reason (R):** The induced current in the closed coil strictly depends upon the rate of change of magnetic flux as well as the ohmic resistance of the coil. [Practice] | **[Chapter 6: Electromagnetic Induction | Confidence: High]**

Q16. Assertion (A): A spherical air bubble trapped inside water generally acts as a concave (diverging) lens. **Reason (R):** The absolute refractive index of air is strictly less than that of water. [PYQ 2018] | **[Chapter 9: Ray Optics | Confidence: Very High]**

Section B: 2-Mark Questions (Very Short Answer)

Q17. Two point charges $+q$ and $-2q$ are fixed at a distance d apart in a vacuum. Carefully draw the continuous electric field lines representing the electrostatic field of this specific system. [PYQ 2017] | **[Chapter 1: Electric Charges & Fields | Confidence: Very High]**

Q18. Three ideal capacitors of capacitances $C_1 = 2 \mu\text{F}$, $C_2 = 3 \mu\text{F}$, and $C_3 = 6 \mu\text{F}$ are connected in series. Find the equivalent capacitance of this combination. If a 10 V battery is connected across this series combination, what is the charge stored on the $3 \mu\text{F}$ capacitor? [PYQ 2015] | **[Chapter 2: Electrostatic Potential & Capacitance | Confidence: High]**

Q19. State Ampere's circuital law mathematically. Use it to briefly determine the magnetic field inside a long, straight, uniform solid current-carrying wire of radius R at a perpendicular distance $r < R$ from its central axis. [PYQ 2014, 2019] | **[Chapter 4: Moving Charges & Magnetism | Confidence: Very High]**

Q20. State two defining characteristics of electromagnetic waves. What is the exact phase difference between the oscillating electric field and magnetic field vectors in an EM wave? [PYQ 2016, 2022] | **[Chapter 8: Electromagnetic Waves | Confidence: Very High]**

Q21. Two photosensitive metals A and B have respective work functions of 2.0 eV and 4.0 eV . Which of the two metals possesses a lower threshold wavelength? Find the explicit mathematical ratio of their threshold wavelengths ($\lambda_{0A} : \lambda_{0B}$). [PYQ 2017] | **[Chapter 11: Dual Nature of Radiation & Matter | Confidence: Very High]**

Section C: 3-Mark Questions (Short Answer)

Q22. Using Gauss's theorem, carefully derive the expression for the electric field intensity at any point due to two infinitely large, thin, parallel plane sheets of charge bearing uniform surface charge densities $+\sigma$ and $-\sigma$ respectively. Evaluate the field strictly in the regions: (i) outside the sheets, and (ii) in between the sheets. [PYQ 2020] | **[Chapter 1: Electric Charges & Fields | Confidence: Very High]**

Q23. In a given electrical circuit, two batteries of emfs $E_1 = 2 \text{ V}$ and $E_2 = 1 \text{ V}$ with respective internal resistances $r_1 = 1 \Omega$ and $r_2 = 1 \Omega$ are connected in parallel, with their positive terminals meeting at a common junction. This combination is connected across an external resistance of $R = 2 \Omega$. Using Kirchhoff's laws, calculate the net current flowing through the external resistor R . [PYQ 2016, 2018] | **[Chapter 3: Current Electricity | Confidence: High]**

Q24. A moving coil galvanometer of internal resistance $G = 50 \Omega$ gives a full-scale deflection for a current $I_g = 2 \text{ mA}$. Calculate the exact magnitude of the necessary resistance required to convert this galvanometer into: (A) an ammeter of range 0 to 5 A . (B) a voltmeter of range 0 to 10 V . [PYQ 2022] | **[Chapter 4: Moving Charges & Magnetism | Confidence: Very High]**

Q25. Compare the inherent magnetic properties of diamagnetic, paramagnetic, and ferromagnetic substances specifically on the basis of their: (i) Relative magnetic permeability (μ_r) (ii) Magnetic susceptibility (χ) (iii) Behaviour when placed in a non-uniform external magnetic field. [PYQ 2019, 2023] | **[Chapter 5: Magnetism & Matter | Confidence: Very High]**

Q26. A rectangular conducting loop of length l and breadth b is pulled out of a uniform transverse magnetic field B with a constant velocity v . Derive the mathematical expressions for: (A) The motional emf induced in the loop. (B) The mechanical force required to pull the loop out. (C) The mechanical power dissipated as Joule heating. [PYQ 2015] | **[Chapter 6: Electromagnetic Induction | Confidence: High]**

Q27. An alternating voltage of 220 V (rms) and frequency 50 Hz is connected to a series combination of a resistor $R = 100\ \Omega$ and a pure capacitor $C = 15\ \mu\text{F}$. Calculate the: (A) Net impedance of the circuit. (B) rms current flowing in the circuit. (C) Voltage drop strictly across the resistor and across the capacitor. [PYQ 2014] | **[Chapter 7: Alternating Current | Confidence: Very High]**

Q28. Describe the single slit diffraction experiment. Draw the graphical pattern showing the spatial variation of intensity with angle in a single slit diffraction. Mathematically show that the central maximum is exactly twice as wide as any of the secondary maxima. [PYQ 2020] | **[Chapter 10: Wave Optics | Confidence: Very High]**

Section D: 4-Mark Questions (Case-Study Based)

Q29. Case Study: Electrostatic Shielding In a thunderstorm accompanied by lightning, it is fundamentally safer to be sitting completely inside a closed car rather than standing under a tree or in the open. This phenomenon is based on the principle of electrostatic shielding. When an external electric field is applied to a metallic conductor, the free electrons redistribute themselves almost instantaneously on the surface such that the net electric field strictly inside the bulk of the conductor and within any hollow cavity inside it becomes exactly zero. Consequently, the entire conductor surface and its interior become an equipotential volume. Sensitive electronic instruments are frequently enclosed in metal boxes to protect them from external electrical disturbances. (A) What is the exact magnitude of the electric field inside a hollow metallic cavity when the conductor is placed in an intense external electric field? (1 Mark) (B) How does the electrostatic potential vary from the surface to the centre of a charged solid metallic sphere? (1 Mark) (C) Two thin hollow concentric conducting spheres of radii r and R ($r < R$) carry charges q and Q respectively. Find the formal expression for the electric potential at a point on the surface of the inner sphere. (2 Marks) [PYQ 2022 / Practice] | **[Chapter 2: Electrostatic Potential & Capacitance | Confidence: Very High]**

Q30. Case Study: Dispersion of Light by a Glass Prism When a narrow beam of white light passes through a glass prism, it splits into its constituent seven colours. This specific phenomenon is called the dispersion of light. Sir Isaac Newton first discovered that white light consists of multiple colours. Dispersion occurs because the refractive index of glass is intrinsically different for different colours (wavelengths) of light. Therefore, each colour bends by a completely different angle as it passes through the prism. Violet light, having the shortest visible wavelength, suffers the maximum deviation, while red light, having the longest visible wavelength, undergoes the minimum deviation. (A) State the exact relation

between the refractive index (μ) of a transparent medium and the wavelength (λ) of light traversing it. (1 Mark) (B) Does a parallel-sided glass slab also produce dispersion of a white light beam? Give a concise reason. (1 Mark) (C) A monochromatic ray of light undergoes minimum deviation $\delta_m = 30^\circ$ while passing through an equilateral glass prism (Angle of prism $A = 60^\circ$). Calculate the refractive index of the material of the prism. (2 Marks) [PYQ 2017, 2020] | **[Chapter 9: Ray Optics | Confidence: Very High]**

Section E: 5-Mark Questions (Long Answer)

Q31. (A) Define the term 'electric dipole moment'. Is it a scalar or a vector quantity? Write its SI unit. (B) Systematically derive the mathematical expression for the electric field intensity $\vec{E}$ at a point on the axial line of an electric dipole. (C) An electric dipole consisting of charges $+q$ and $-q$ separated by a distance $2a$ is placed in a uniform external electric field $\vec{E}$. Derive the mathematical expression for the potential energy of the dipole in this field. (D) Under what exact orientation (angle θ) is the potential energy of the dipole considered to be perfectly zero? [PYQ 2019, 2023] | **[Chapter 1: Electric Charges & Fields / Chapter 2: Potential | Confidence: Very High]**

Q32. (A) Define the terms 'Self-Inductance' and 'Mutual Inductance'. Write their common SI unit. (B) Derive the mathematical expression for the mutual inductance (M) of two long, tightly wound, coaxial solenoids of the same length l . Let the inner solenoid have N_1 turns and radius r_1 , and the outer solenoid have N_2 turns and radius r_2 . (C) A pair of closely adjacent coils has a known mutual inductance of 1.5 H . If the steady current in the primary coil drops from 10 A to exactly zero in 0.2 s , calculate the magnitude of the average emf induced in the secondary coil. [PYQ 2015, 2022] | **[Chapter 6: Electromagnetic Induction | Confidence: Very High]**

Q33. (A) Draw a neat, fully labelled ray diagram showing the image formation by an astronomical refracting telescope when the final image is formed entirely at the least distance of distinct vision (D). (B) Using this ray diagram, logically derive the expression for the magnifying power (m) of the astronomical telescope in this specific adjustment. (C) An astronomical telescope uses an objective lens of focal length 144 cm and an eyepiece of focal length 6.0 cm . Calculate its exact magnifying power and the physical separation between the objective and the eyepiece when it is in normal adjustment (image at infinity). [PYQ 2014, 2018] | **[Chapter 9: Ray Optics | Confidence: Very High]**

Answer Key

Q1: (a) $1/2$. By Coulomb's law, $F \propto q(Q - q) = qQ - q^2$. For max force, $dF/dq = Q - 2q = 0 \Rightarrow q = Q/2$. Ratio $q/Q = 1/2$. **Q2:** (c) Zero. The electric field is perpendicular to the plates. Moving parallel to the plates means moving along an equipotential surface, so $\Delta V = 0$ and Work = $q\Delta V = 0$. **Q3:** (d) ρ . Resistivity is an intrinsic

property of the material and strongly depends on temperature, not on the physical dimensions (length or area) of the wire. **Q4:** (b) $n^2 B$. The new radius becomes $r' = r/n$. The magnetic field for 1 turn is $B = \mu_0 I / 2r$. For n turns, $B' = n\mu_0 I / 2r' = n\mu_0 I / 2(r/n) = n^2(\mu_0 I / 2r) = n^2 B$. **Q5:** (b) from the stronger part of the field to the weaker part. Diamagnetic materials are feebly repelled by magnets. **Q6:** (c) strictly less than g . As the coin falls, the changing magnetic flux induces eddy currents in it. By Lenz's law, these currents produce an upward magnetic force that opposes gravity. **Q7:** (c) 0. For a pure capacitor, phase angle $\phi = 90^\circ$. Power factor $\cos(90^\circ) = 0$. **Q8:** (c) the speed of light in a vacuum (c). By Maxwell's equations, $E_0/B_0 = c$. **Q9:** (c) -25 cm , Concave. $f = 1/P = 1/-4.0 = -0.25 \text{ m} = -25 \text{ cm}$. A negative focal length strictly denotes a concave (diverging) lens. **Q10:** (b) λ/a . The condition for minima in single slit diffraction is $a \sin \theta = n\lambda$. For the first minimum ($n = 1$), $\sin \theta \approx \theta = \lambda/a$. **Q11:** (c) Planck's constant (h). By Einstein's equation, $K_{max} = h\nu - \Phi$. This is of the form $y = mx + c$. Thus, the slope $m = h$. **Q12:** (b) $1/2$ of its initial value. $\lambda = h/\sqrt{2mK}$. If K is multiplied by 4, $\sqrt{4} = 2$. So $\lambda' = \lambda/2$.

Q13: (a) Both A and R are true and R is the correct explanation of A. **Q14:** (a) Both A and R are true and R is the correct explanation of A. ($V = E - Ir$). **Q15:** (b) Both A and R are true but R is NOT the correct explanation of A. EMF depends *only* on the rate of flux change (Faraday's law), while current $I = \mathcal{E}/R$ depends on resistance. **Q16:** (a) Both A and R are true and R is the correct explanation of A. When a lens of lower refractive index is placed in a medium of higher refractive index, its nature reverses (convex acts as concave).

Q17: Draw a $+q$ and a $-2q$ charge. Electric field lines strictly originate from $+q$ and terminate on $-2q$. Because $-2q$ has double the magnitude, exactly twice as many lines should terminate on $-2q$ as originate from $+q$ (some lines come from infinity to $-2q$). **Q18:** Series: $1/C_{eq} = 1/2 + 1/3 + 1/6 = 3/6 + 2/6 + 1/6 = 6/6 = 1 \Rightarrow C_{eq} = 1 \mu\text{F}$. Charge in series is identical for all capacitors: $Q = C_{eq} \times V = 1 \mu\text{F} \times 10 \text{ V} = 10 \mu\text{C}$. **Q19:** $\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{enclosed}$. For a point inside ($r < R$), the Amperian loop encloses current $I' = I(r^2/R^2)$. $B(2\pi r) = \mu_0 I r^2 / R^2 \Rightarrow B = (\mu_0 I r) / (2\pi R^2)$. **Q20:** (1) EM waves are transverse in nature and do not require a material medium. (2) The phase difference between oscillating $\vec{E}$ and $\vec{B}$ vectors is exactly 0 (they are perfectly in phase). **Q21:** Metal B has a lower threshold wavelength. $\Phi = hc/\lambda_0 \Rightarrow \lambda_0 \propto 1/\Phi$. Ratio $\lambda_{0A}/\lambda_{0B} = \Phi_B/\Phi_A = 4.0/2.0 = 2 : 1$.

Q22: Let Sheets 1 and 2 have $+\sigma$ and $-\sigma$. Field due to a single sheet is $E = \sigma/2\epsilon_0$. (i) Outside (left of 1 or right of 2): Fields are opposite and equal, so $E_{net} = \sigma/2\epsilon_0 - \sigma/2\epsilon_0 = 0$. (ii) Between the sheets: Fields from both point in the same direction (+ to -), so $E_{net} = \sigma/2\epsilon_0 + \sigma/2\epsilon_0 = \sigma/\epsilon_0$. **Q23:** Let the junction of positive terminals be A. Current I_1 from E_1 , I_2 from E_2 . Net current through R is $I = I_1 + I_2$. Loop 1 (E_1): $2 - 1(I_1) - 2(I_1 + I_2) = 0 \Rightarrow 3I_1 + 2I_2 = 2$. Loop 2 (E_2): $1 - 1(I_2) - 2(I_1 + I_2) = 0 \Rightarrow 2I_1 + 3I_2 = 1$. Solving: $I_1 = 4/5 \text{ A}$, $I_2 = -1/5 \text{ A}$. Net current $I = 4/5 - 1/5 = 3/5 = 0.6 \text{ A}$. **Q24:** (A) Ammeter: Shunt resistor S in parallel. $S = I_g G / (I - I_g) = (2 \times 10^{-3} \times 50) / (5 - 0.002) \approx 0.1/5 = 0.02 \Omega$. (B) Voltmeter:

High resistor R_s in series.

$R_s = V/I_g - G = 10/(2 \times 10^{-3}) - 50 = 5000 - 50 = 4950 \Omega$. **Q25:** (i) μ_r : Dia ($0 \leq \mu_r < 1$), Para ($\mu_r > 1$, slightly), Ferro ($\mu_r \gg 1$, very large). (ii) χ : Dia (small, negative), Para (small, positive), Ferro (large, positive). (iii) Non-uniform field: Dia slowly moves to weaker field, Para slowly moves to stronger field, Ferro rapidly moves to stronger field. **Q26:** (A) Rate of change of area $dA/dt = -l \cdot dx/dt = -lv$. $\epsilon = -d\Phi/dt = -d(BA)/dt = Blv$. (B) Current $I = \epsilon/R = Blv/R$. Magnetic force opposing motion $F_m = IlB = (Blv/R)lB = B^2l^2v/R$. Mechanical force $F_{ext} = F_m = B^2l^2v/R$. (C) Power $P = F_{ext}v = B^2l^2v^2/R$. **Q27:** (A)

$X_C = 1/(2\pi fC) = 1/(2 \times 3.14 \times 50 \times 15 \times 10^{-6}) \approx 212.3 \Omega$. Impedance

$$Z = \sqrt{R^2 + X_C^2} = \sqrt{100^2 + 212.3^2} \approx 234.7 \Omega. \text{ (B)}$$

$I_{rms} = V_{rms}/Z = 220/234.7 \approx 0.937 \text{ A}$. (C) $V_R = I_{rms}R = 0.937 \times 100 = 93.7 \text{ V}$.

$V_C = I_{rms}X_C = 0.937 \times 212.3 \approx 198.9 \text{ V}$. **Q28:** Draw intensity graph with maximum at centre, decaying secondary peaks. For central max, angular spread is between 1st minima on either side: $\theta = \lambda/a$ to $-\lambda/a$, so width $= 2\lambda/a$. For secondary max (e.g., between 1st and 2nd minimum), angular width is $2\lambda/a - \lambda/a = \lambda/a$. Hence central max ($2\lambda/a$) is twice as wide as secondary max (λ/a).

Q29: (A) Exactly zero. (B) The potential is completely uniform and strictly constant from the surface to the centre (Equipotential). (C) Total potential at inner sphere surface

$$V_{inner} = V_{due_to_q} + V_{due_to_Q} = \frac{1}{4\pi\epsilon_0} \frac{q}{r} + \frac{1}{4\pi\epsilon_0} \frac{Q}{R} = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{r} + \frac{Q}{R} \right).$$

Q30: (A) Cauchy's formula (or general inverse relation): $\mu = A + B/\lambda^2$. Refractive index is inversely proportional to the square of wavelength. (B) No. A parallel-sided slab causes lateral displacement but emergent rays of all colours emerge perfectly parallel to the incident ray, so they overlap and recombine, preventing visible dispersion at macroscopic distances. (C)

$$\mu = \frac{\sin((A+\delta_m)/2)}{\sin(A/2)} = \frac{\sin((60^\circ+30^\circ)/2)}{\sin(60^\circ/2)} = \frac{\sin 45^\circ}{\sin 30^\circ} = \frac{1/\sqrt{2}}{1/2} = \frac{2}{\sqrt{2}} = \sqrt{2} \approx 1.414.$$

Q31: (A) Product of the magnitude of either charge and the vector distance between them ($\vec{p} = q \times 2\vec{a}$). Vector quantity directed from negative to positive charge. Unit: Coulomb-metre

(C m). (B) Use superposition: $E = E_{+q} - E_{-q} = \frac{q}{4\pi\epsilon_0(r-a)^2} - \frac{q}{4\pi\epsilon_0(r+a)^2}$. Simplify to get

$$E = \frac{1}{4\pi\epsilon_0} \frac{2pr}{(\tau^2 - a^2)^2}. \text{ (C) Work done } dW = \tau d\theta = pE \sin \theta d\theta. \text{ Integrate from } 90^\circ \text{ to } \theta:$$

$$U = \int_{90^\circ}^{\theta} pE \sin \theta d\theta = -pE \cos \theta = -\vec{p} \cdot \vec{E}. \text{ (D) Potential energy is exactly zero when the dipole is strictly perpendicular to the electric field } (\theta = 90^\circ).$$

Q32: (A) Self-inductance:

Phenomenon where a changing current in a coil induces an emf in itself. Mutual: A changing current in one coil induces an emf in a neighbouring coil. Unit: Henry (H). (B) Flux in inner coil

$$\Phi_1 = N_1(B_2 \cdot A_1) = N_1(\mu_0 n_2 I_2)(\pi r_1^2). M = \Phi_1/I_2 = \mu_0 N_1(N_2/l)\pi r_1^2 = \frac{\mu_0 N_1 N_2 \pi r_1^2}{l}. \text{ (C)}$$

Induced emf $e = M|dI/dt| = 1.5 \times |(0 - 10)/0.2| = 1.5 \times (10/0.2) = 1.5 \times 50 = 75 \text{ V}$

Q33: (A) Draw parallel rays entering objective, forming real inverted image at F_o . Place eyepiece such that this image lies strictly between optical centre and F_e , yielding magnified virtual final image at D . (B) Magnifying power $m = \tan \beta / \tan \alpha \approx \beta / \alpha$. $\alpha \approx h'/f_o$ and $\beta \approx h'/u_e$. So $m = f_o/u_e$. Apply lens formula for eyepiece ($v_e = -D$) to get

$1/u_e = 1/f_e + 1/D$. Substitute to yield $m = \frac{f_o}{f_e} \left(1 + \frac{f_e}{D}\right)$. (C) For normal adjustment (infinity), $m = f_o/f_e = 144/6.0 = 24$. Separation (Tube length)
 $L = f_o + f_e = 144 + 6.0 = 150 \text{ cm}$.

CBSE Class 12 (2027) — Half-Yearly Predicted Paper (80% syllabus). AI-generated via PYQ/Blueprint analysis, not actual exam questions.